

AI Compiler Performance Engineering

Auto-Loops Books — ch03

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Chapter 1

AI Hardware Architecture and Compiler Constraints

1.1 Hardware Landscape

1.2 Constraint Matrix

1.3 Gpu Constraints

1.4 Cpu Constraints

1.5 Npu Constraints

1.6 Hardware Aware Compile

1.7 Benchmark Method

1.8 Yirage Modeling

Table 1.1: AI accelerator classes and primary execution models.

Class	Execution model	Decode fit
NVIDIA GPU	SIMT + tensor cores	Graphs, fusion
AMD GPU	Wavefront + matrix cores	HIP fusion
x86/ARM CPU	Cache + SIMD	Coarse fusion
Edge NPU	Spatial dataflow	Static megakernel

Table 1.2: Cross-hardware constraint matrix.

Dimension	GPU	CPU	NPU
On-chip SRAM	Shared mem / L1	L1/L2/L3	Tile buffers
Parallel grain	Warp	Core + SIMD	Spatial PE array
Dynamic shapes	Flexible	Flexible	Restricted

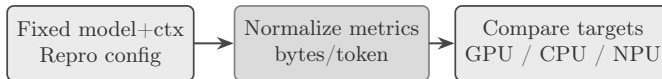


Figure 1.1: Cross-hardware benchmark methodology used in this book.

Bibliography